Alzheimer’s Disease Prediction System Using Machine Learning Classification

A Project Report Submitted in Partial Fulfilment of the Requirements for the Course

DATA SCIENCE AND ARTIFICIAL INTELLIGENCE

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Certificate

This is to certify that the project entitled “Alzheimer’s Disease Prediction System Using Machine Learning Classification” is a bona fide work carried out by the students named above under my supervision and guidance in partial fulfilment of the requirements for the award of the degree of [DEGREE NAME].

Project Guide: _____ Head of Department: _____

Date: Place:

Declaration

We hereby declare that the project entitled “Alzheimer’s Disease Prediction System Using Machine Learning Classification” is our original work carried out under the guidance of Dr. Suresan Pareth. This work has not been submitted elsewhere for the award of any degree or diploma, except where acknowledged in the report.

Place: Signature of Students:

Date:

Acknowledgement

We express our sincere gratitude to our project guide, Dr. Suresan Pareth, the Head of the Department, the faculty members, and our institution for their valuable guidance, support, and encouragement throughout this project. We also thank everyone who contributed to the successful completion of this work.

Abstract

Alzheimer's disease is a progressive neurological disorder and the most common cause of dementia. Early identification of at-risk individuals supports timely clinical attention and care planning. This project proposes a data-driven system that analyzes patient records and predicts whether a patient has Alzheimer's disease.

The dataset contains 2,149 patient records with 32 predictive attributes covering demographics, lifestyle, medical history, clinical measurements, cognitive and functional assessments, and symptoms. Statistical techniques are used to study the distributions of the attributes and their correlation with the diagnosis. Preprocessing checks for missing values, duplicates and inconsistent measurements, and standardizes features for distance-based models.

Five supervised classification algorithms are compared: K-Nearest Neighbours, Decision Tree, Support Vector Machine, Random Forest and Gradient Boosting. Gradient Boosting achieved the best test ROC-AUC (0.946) with an accuracy of 94.7%, precision of 92.7%, recall of 92.1% and F1-score of 92.4%. Visualizations show that cognitive and functional scores (MMSE, Functional Assessment and ADL) and memory complaints are the strongest indicators.

The results demonstrate how Data Science and AI can support early screening. The system is an educational prototype and not a clinical diagnostic tool.

Keywords: Alzheimer's Disease, Data Science, Machine Learning, Classification, Statistical Analysis, Prediction.

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1 Introduction

1.1 Background

Alzheimer’s disease gradually damages memory, thinking and the ability to carry out daily activities, and it is the most common cause of dementia in older adults. There is no simple single test for it; clinicians combine patient history, cognitive tests such as the Mini-Mental State Examination (MMSE), functional assessments and lifestyle and medical information. Early detection allows earlier treatment, better care planning and support for families. Data-driven models can help clinicians screen large numbers of patients by finding patterns that are hard to see manually.

1.2 Problem Statement

Conventional assessment depends on manual evaluation of many clinical, cognitive and lifestyle factors, which is time-consuming and can vary between examiners. There is a need for a data-driven method that analyzes these factors together and predicts the likelihood of Alzheimer's disease quickly and consistently, while showing which factors drive the prediction.

1.3 Objectives

1. To analyze patient attributes using statistical techniques.
2. To preprocess the Alzheimer's dataset.
3. To investigate correlations between the attributes and the diagnosis.
4. To develop a machine learning model for Alzheimer's classification.
5. To visualize and interpret the results.

1.4 Scope of the Project

The project uses a publicly available tabular dataset of 2,149 patients aged 60 to 90. It covers 32 attributes spanning demographics, lifestyle, medical history, clinical measurements, cognitive and functional assessments and symptoms. The intended application is research and learning in early screening support. The work does not cover medical images and is not intended for real clinical decisions.

2 Dataset Description and Representation

2.1 Data Collection

The data comes from a public Alzheimer's disease dataset stored in the file alzheimers_disease_data.csv (the exact source and URL are listed in the Bibliography). It has 2,149 observations and 35 columns: 32 predictive features, the target column Diagnosis, an identifier column PatientID and a column DoctorInCharge. The collection period is not stated in the dataset. There are no missing values and no duplicate records. Patient ages range from 60 to 90, and gender is almost balanced (1,088 and 1,061 records for the two codes). The target is imbalanced: 1,389 patients (64.6%) have no Alzheimer's disease and 760 patients (35.4%) have it.

2.2 Dataset Features

Table 2.1: Alzheimer's dataset features

Group	Feature	Type	Description
Demographic	Age	Numerical	Age of the patient (60–90)
Demographic	Gender, Ethnicity, EducationLevel	Categorical (coded)	Gender, ethnic group and education level codes
Lifestyle	BMI	Numerical	Body mass index
Lifestyle	Smoking, AlcoholConsumption	Binary / Numerical	Smoking status and alcohol use
Lifestyle	PhysicalActivity, DietQuality, SleepQuality	Numerical	Activity, diet and sleep quality scores
Medical history	FamilyHistoryAlzheimers, CardiovascularDisease, Diabetes, Depression, HeadInjury, Hypertension	Binary	0 = no, 1 = yes
Clinical	SystolicBP, DiastolicBP	Numerical	Blood pressure
Clinical	CholesterolTotal, CholesterolLDL, CholesterolHDL, CholesterolTriglycerides	Numerical	Cholesterol measurements
Cognitive / functional	MMSE	Numerical	Mini-Mental State Examination score (0–30, lower is worse)
Cognitive / functional	FunctionalAssessment, ADL	Numerical	Functional and daily-living scores (0–10, lower is worse)
Cognitive / functional	MemoryComplaints, BehavioralProblems	Binary	Reported memory complaints and behavioral problems

Group	Feature	Type	Description
Symptoms	Confusion, Disorientation, PersonalityChanges, DifficultyCompletingTasks, Forgetfulness	Binary	Presence of each symptom
Target	Diagnosis	Categorical	0 = no Alzheimer's disease, 1 = Alzheimer's disease

2.3 Tabular Representation

The dataset is a table in which each row is one patient and each column is one measured feature. For example, one row holds the Age, BMI, MMSE, ADL and other values of a single patient together with that patient's Diagnosis. The columns PatientID and DoctorInCharge are not used for modelling.

2.4 Matrix Representation

The predictive part of the dataset is represented by a matrix X with m = 2,149 samples (rows) and n = 32 features (columns), and a target vector y:

$$X = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \dots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{bmatrix} \qquad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}$$

Here x_ij is the value of feature j for patient i, and y_i is 0 (no Alzheimer's disease) or 1 (Alzheimer's disease).

3 Data Preprocessing and Statistical Analysis

3.1 Data Cleaning

- **Missing values:** the dataset contains none, so no imputation was needed.
- **Duplicate records:** none were found.
- **Inconsistent measurements:** values were checked against valid ranges (for example MMSE between 0 and 30, and the 0–10 scores between 0 and 10). All values were consistent.

- **Outliers:** box plots and summary statistics showed all values inside plausible ranges, so no records were removed.
- **Irrelevant columns:** PatientID is only an identifier and DoctorInCharge holds one constant value, so both were dropped before modelling.

3.2 Data Transformation and Normalization

Distance-based models (K-Nearest Neighbours and SVM) are sensitive to feature scale, so features were standardized using z-score normalization:

$z = (x - \mu) / \sigma$

where μ is the mean and σ the standard deviation of the feature in the training data. Min-Max normalization, $x' = (x - x_{min}) / (x_{max} - x_{min})$, is an alternative that rescales values to the range 0 to 1. Tree-based models (Decision Tree, Random Forest, Gradient Boosting) do not depend on scale and use the original values.

3.3 Descriptive Statistics

Table 3.1: Descriptive statistics of key features (n = 2,149)

Feature	Mean	Median	Mode*	Variance	Std. Dev.
Age	74.91	75.00	68	80.82	8.99
BMI	27.66	27.82	37	52.09	7.22
MMSE	14.76	14.44	6	74.19	8.61
FunctionalAssessment	5.08	5.09	9	8.37	2.89
ADL	4.98	5.04	9	8.70	2.95
SleepQuality	7.05	7.12	8	3.11	1.76
DietQuality	4.99	5.08	8	8.46	2.91
PhysicalActivity	4.92	4.77	3	8.16	2.86
CholesterolTotal	225.20	225.09	215	1809.84	42.54

*Mode of the values rounded to the nearest integer, because the features are continuous.

The mean and median of every feature are very close, so the distributions are roughly symmetric. MMSE has a wide spread (standard deviation 8.61 on a 0–30 scale), which gives it room to separate patients with and without the disease. Comparing the two

groups, patients with Alzheimer’s disease have a lower average MMSE (11.99 against 16.27), Functional Assessment (3.65 against 5.86) and ADL (3.66 against 5.71). Average age is almost identical in the two groups (74.84 against 74.95).

3.4 Correlation Analysis

The Pearson correlation between two variables X and Y is

$r_{XY} = \mathrm{Cov}(X, Y) / (\sigma_X \sigma_Y)$

Table 3.2: Features most correlated with the diagnosis

Feature	Correlation with Diagnosis
FunctionalAssessment	−0.365
ADL	−0.332
MemoryComplaints	+0.307
MMSE	−0.237
BehavioralProblems	+0.224
SleepQuality	−0.057
EducationLevel	−0.044
CholesterolHDL	+0.043
Hypertension	+0.035
FamilyHistoryAlzheimers	−0.033

Five features stand out. Lower functional, daily-living and MMSE scores are associated with the disease (negative correlations), while memory complaints and behavioral problems are associated with it positively. All other features have weak correlations (absolute value below 0.06). The correlations between the cognitive and functional scores themselves are close to zero (for example MMSE and ADL: 0.00), which means each score contributes separate information.

4 Proposed Methodology

4.1 System Workflow

1. Data collection: load alzheimers_disease_data.csv.
2. Data preprocessing: check quality, drop identifier columns, standardize features where needed.
3. Statistical analysis: descriptive statistics and correlations.
4. Feature selection: use all 32 clinical features.
5. Model development: train five classification algorithms.
6. Model evaluation: cross-validation and a held-out test set.
7. Visualization and interpretation: charts, feature importance and a prediction function for new patients.

4.2 Machine Learning Algorithm

The following supervised classification algorithms were compared:

- **K-Nearest Neighbours (KNN):** classifies a patient by the majority class of the 7 most similar training patients.
- **Decision Tree:** a sequence of if/else splits on feature values (maximum depth 6).
- **Support Vector Machine (SVM):** finds a boundary with the widest margin between the classes using an RBF kernel; probabilities are obtained with probability calibration.
- **Random Forest:** 300 decision trees trained on random samples of the data, voting together.
- **Gradient Boosting:** the selected model, described below.

Gradient Boosting builds many shallow decision trees one after another. Each new tree is fitted to the errors (the gradient of the loss function) left by the trees before it, and the sum of all trees gives a score that is converted into a probability of Alzheimer’s disease. A patient is classified as positive when this probability is at least 0.5. Boosting reduces both bias and variance and usually performs very well on tabular data.

4.3 Feature Engineering

All 32 original features were used. PatientID and DoctorInCharge were removed because they carry no clinical information. Features were scaled only for KNN and SVM. No new features were created. Feature importance from the tree models was used afterwards to interpret which attributes drive the predictions.

4.4 Model Training and Testing

The data was split into 80% training (1,719 patients) and 20% testing (430 patients) using a stratified split so that both parts keep the same class ratio (random state 42). Each model was first assessed with 5-fold stratified cross-validation on the training set using ROC-AUC, then trained on the full training set and evaluated once on the test set.

5 Implementation

5.1 Software and Tools

- Programming language: Python 3, executed in JupyterLab.
- Libraries: NumPy, Pandas, Matplotlib, Seaborn.
- Machine learning: Scikit-learn; model saving with Joblib.
- AI assistance: Claude (Anthropic).

5.2 Implementation Procedure

1. Install and import the libraries.
2. Load the CSV file into a Pandas DataFrame.
3. Explore the data: shape, missing values, duplicates, class balance, summary statistics.
4. Plot the correlation chart and the box plots.
5. Separate the features X and the target y, and split into training and test sets.
6. Compare the five models with cross-validation.
7. Train the models and evaluate them on the test set.
8. Plot the confusion matrix, ROC curves and feature importances.
9. Save the best model and use it to predict for new patients.

5.3 Sample Code

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import GradientBoostingClassifier
from sklearn.metrics import classification_report

df = pd.read_csv('alzheimers_disease_data.csv')
print(df.shape)
print(df.isna().sum().sum(), df.duplicated().sum())
print(df['Diagnosis'].value_counts())

X = df.drop(columns=['Diagnosis', 'PatientID', 'DoctorInCharge'])
y = df['Diagnosis']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, stratify=y, random_state=42
)

model = GradientBoostingClassifier(random_state=42)
model.fit(X_train, y_train)

pred = model.predict(X_test)
print(classification_report(y_test, pred))
```

The statements above load the data, verify its quality, separate the features from the target, create a stratified 80/20 split, train a Gradient Boosting classifier and report precision, recall and F1-score on the unseen test data. The complete notebook additionally compares all five models, draws the charts and defines the function predict_patient(), which takes a dictionary of feature values, fills any missing feature with the dataset median and returns the predicted class and its probability.

6 Results and Discussion

6.1 Statistical Results

The statistics in Tables 3.1 and 3.2 show that the cognitive and functional scores differ clearly between the two groups, while age, BMI, cholesterol and most lifestyle variables do not. Memory complaints and behavioral problems are the symptom features most linked to the diagnosis.

6.2 Data Visualization

The notebook produces the following plots (insert the figures from the notebook output here):

- **Figure 6.1: Class distribution.** A bar chart showing 1,389 patients without and 760 with Alzheimer's disease, confirming moderate class imbalance.
- **Figure 6.2: Correlation with the diagnosis.** A bar chart ranking all features; FunctionalAssessment, ADL, MemoryComplaints, MMSE and BehavioralProblems are the longest bars.
- **Figure 6.3: Box plots.** MMSE, Functional Assessment and ADL are clearly lower for patients with the disease, while the Age box plots are almost identical.
- **Figure 6.4: Confusion matrix and ROC curves.** The ROC curves of Random Forest and Gradient Boosting lie closest to the top-left corner.
- **Figure 6.5: Feature importance.** Functional Assessment, ADL, MMSE and MemoryComplaints are the most important features.

6.3 Model Evaluation

Accuracy = (TP + TN) / (TP + TN + FP + FN)

Precision = TP / (TP + FP) Recall = TP / (TP + FN) F1 = 2 × Precision × Recall / (Precision + Recall)

Table 6.1: Model comparison (test set of 430 patients)

Model	CV ROC-AUC (mean ± std)	Accuracy	Precision	Recall	F1-score	Test ROC-AUC
Random Forest	0.956 ± 0.013	0.949	0.945	0.908	0.926	0.940
Gradient Boosting	0.954 ± 0.012	0.947	0.927	0.921	0.924	0.946
Decision Tree	0.914 ± 0.015	0.928	0.895	0.901	0.898	0.912
SVM	0.894 ± 0.015	0.835	0.772	0.757	0.764	0.896
KNN	0.770 ± 0.020	0.709	0.645	0.395	0.490	0.765

Gradient Boosting was selected as the final model because it has the highest test ROC-AUC (0.946). Random Forest performs almost identically, so the two are practically tied.

Table 6.2: Confusion matrix of the Gradient Boosting model

	Predicted: No Alzheimer's	Predicted: Alzheimer's
Actual: No Alzheimer's	267 (TN)	11 (FP)
Actual: Alzheimer's	12 (FN)	140 (TP)

Accuracy = (140 + 267) / 430 = 94.7%, precision = 140 / 151 = 92.7%, recall = 140 / 152 = 92.1%.

Table 6.3: Feature importance (Gradient Boosting)

Feature	Importance
MMSE	0.240
FunctionalAssessment	0.239
ADL	0.229
MemoryComplaints	0.155
BehavioralProblems	0.086
DietQuality	0.008
CholesterolLDL	0.006
PhysicalActivity	0.006

The five most important features account for about 95% of the total importance, which agrees with the correlation analysis.

6.4 Discussion

The tree-based ensembles clearly outperform the other models. KNN performs worst, most likely because many of the 32 features carry little information and dilute the distance between patients. SVM was run with default settings and may improve with hyperparameter tuning.

The prediction function behaves sensibly. A patient with low scores (age 78, MMSE 14, Functional Assessment 3, ADL 3, with memory complaints and forgetfulness) receives a probability of Alzheimer's disease of about 99%, while a patient with high scores (age 68, MMSE 24, Functional Assessment 7, ADL 7 and no complaints) receives about 1%.

Limitations. The very high accuracy should be interpreted with care. The features are almost uncorrelated with each other and the dataset appears to be synthetic or heavily curated, so real clinical data would be noisier and the accuracy lower. The model relies heavily on cognitive test scores that clinicians already use to diagnose the disease. No external dataset was used for validation. Finally, 12 of the 152 patients with the disease in the test set were missed (about 8%), which matters in a medical setting.

7 AI Tools and Ethical Considerations

7.1 Use of AI Tools

Claude was used as an AI assistant for:

- Understanding the dataset and its features.
- Code assistance and debugging, including fixing deprecation warnings and making the prediction cell independent of earlier cells.
- Generating analytical summaries of the results.
- Preparing visualizations.
- Preparing this report.

All AI-generated code and text were run, checked against the actual results and edited by the team.

7.2 Ethical Considerations

- **Data reliability:** results depend on the quality and representativeness of the data; a synthetic or biased dataset can give misleadingly high performance.
- **Privacy:** medical data is sensitive. The dataset is anonymized (patient identifiers are not used for modelling and the doctor column is masked); any real deployment would need consent, secure storage and legal compliance.
- **Fairness:** features such as gender and ethnicity should be monitored so that the model does not perform worse for particular groups.
- **Responsible use:** the model is a screening-support prototype. A positive or negative prediction must never replace a clinician's judgment, and false negatives can delay care.
- **Explainability:** feature importance and probabilities are reported so that users can see why a prediction was made.

8 Conclusion and Future Scope

8.1 Conclusion

This project built a classification system that predicts Alzheimer's disease from 32 patient attributes. After checking data quality, analyzing statistics and correlations and comparing five algorithms, Gradient Boosting gave the best result with 94.7% accuracy, 92.7% precision, 92.1% recall and a test ROC-AUC of 0.946 on 430 unseen patients. MMSE, Functional Assessment, ADL, memory complaints and behavioral problems were identified as the key predictors. The results show that Data Science and AI can support early screening, within the limitations noted above.

8.2 Future Scope

- Validation on real, larger and external clinical datasets.
- Hyperparameter tuning and additional models such as XGBoost.
- Model explanation with SHAP values.
- Image-based detection from MRI scans using deep learning (for example EfficientNetB3 transfer learning).
- A web or mobile application where a clinician enters patient values and receives a prediction.
- Longitudinal data to predict disease progression over time.

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